

Minor Project Report

Diabetes Prediction using Machine Learning

Submitted By

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1. Title

Diabetes Prediction using Machine Learning

2. Problem Statement

Diabetes is one of the most common chronic diseases affecting millions of people worldwide. Early detection of diabetes can help healthcare professionals provide timely treatment and reduce the risk of severe complications.

The objective of this project is to develop a supervised machine learning model that predicts whether a patient has diabetes based on demographic and medical attributes such as age, gender, body mass index (BMI), hypertension, heart disease, smoking history, HbA1c level, and blood glucose level.

3. Dataset Description

The dataset used in this project is the **Diabetes Prediction Dataset**, obtained from Kaggle.

Dataset Source:

<https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset>

The dataset initially contained **100,000 patient records** and **9 attributes** describing medical and demographic information.

Features

- Gender
- Age
- Hypertension
- Heart Disease
- Smoking History
- BMI
- HbA1c Level
- Blood Glucose Level

Target Variable

Diabetes

- 0 → Non-Diabetic
- 1 → Diabetic

4. Data Preprocessing

The following preprocessing steps were performed before training the machine learning model:

- Imported the dataset using Pandas.
- Checked the dataset structure using `head()`, `info()`, and `describe()`.
- Verified that there were no missing values.
- Removed duplicate records from the dataset.
- Applied One-Hot Encoding to categorical variables (Gender and Smoking History).
- Split the dataset into training and testing sets using an 80:20 ratio.
- Applied StandardScaler to normalize the numerical features.

These preprocessing steps ensured that the dataset was clean and suitable for machine learning.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics of the dataset before training the model.

The following visualizations were created:

- Distribution of Diabetes

- Gender Distribution
- Smoking History Distribution
- Age Distribution
- BMI Distribution
- Blood Glucose Level Distribution
- HbA1c Level Distribution
- Age vs Diabetes
- BMI vs Diabetes
- Blood Glucose Level vs Diabetes
- HbA1c Level vs Diabetes
- Correlation Heatmap

Key Observations

- The dataset is imbalanced, with a higher number of non-diabetic patients than diabetic patients.
- Older individuals showed a higher likelihood of having diabetes.
- Higher BMI values were more common among diabetic patients.
- Blood glucose level and HbA1c level showed a strong positive relationship with diabetes.
- Hypertension and heart disease also contributed positively toward diabetes prediction.

(Insert all EDA graphs here with proper captions.)

6. Model Used

The machine learning algorithm used in this project is:

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. Since the target variable contains only two classes (Diabetic and Non-Diabetic), Logistic Regression is an appropriate choice.

The model was trained using the training dataset and later evaluated using the testing dataset.

7. Training Process

The complete workflow followed in this project is shown below:

1. Import Libraries
2. Load Dataset
3. Dataset Understanding
4. Data Cleaning
5. Exploratory Data Analysis

6. Feature Encoding
7. Train-Test Split
8. Feature Scaling
9. Logistic Regression Model Training
10. Model Evaluation
11. Save Trained Model using Pickle

The trained model was saved as **diabetes_model.pkl**, while the StandardScaler object was saved as **scaler.pkl**.

8. Evaluation Metrics and Results

The Logistic Regression model was evaluated using standard classification metrics.

Metric	Value
Accuracy	95.71%
Precision	85.44%
Recall	62.75%
F1 Score	72.36%

Confusion Matrix

(Insert the confusion matrix image here.)

Result Analysis

The Logistic Regression model achieved an overall accuracy of **95.71%**. The high precision indicates that when the model predicts a patient as diabetic, it is usually correct. The recall score is comparatively lower due to the imbalance in the dataset, where diabetic cases are significantly fewer than non-diabetic cases. Overall, the model demonstrates good performance for diabetes prediction and provides reliable classification results.

9. Conclusion

In this project, a supervised machine learning model was developed to predict diabetes using demographic and medical information.

The dataset was successfully cleaned, preprocessed, and analyzed using various visualization techniques. Logistic Regression was used as the classification algorithm and achieved an accuracy of **95.71%**.

The trained model can effectively classify diabetic and non-diabetic patients and can serve as a baseline model for future improvements. In future work, advanced machine learning algorithms, hyperparameter tuning, and techniques for handling class imbalance can be explored to further improve prediction performance.

10. References

1. Kaggle Diabetes Prediction Dataset
<https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset>
2. Scikit-learn Documentation
<https://scikit-learn.org/>
3. Pandas Documentation
<https://pandas.pydata.org/>
4. NumPy Documentation
<https://numpy.org/>
5. Matplotlib Documentation
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